A cross-platform–replicated activated-fibroblast signature in ligamentum flavum hypertrophy nominates locally-deliverable anti-fibrotic repurposing candidates

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Abstract

Ligamentum flavum (LF) hypertrophy is a fibrotic driver of degenerative lumbar spinal stenosis for which there is no approved pharmacotherapy. Using public human single-cell RNA-seq of LF (GSE294458), we derived a within-tissue *activated-fibroblast* transcriptional program (a matrix-depositing state contrast) and tested its reproducibility in two independent LF cohorts: a second single-cell cohort (GSE267819; three hypertrophic donors) and a bulk cohort profiled on a different platform (GSE113212, Agilent microarray). The signature replicated in both: all 22 tested genes were concordant in the independent single-cell cohort (rank-enrichment $p = 9.4\times10^{-16}$) and 18 of 20 in the bulk cohort ($p = 1.5\times10^{-7}$), led by fibrillar collagens, small leucine-rich proteoglycans, HTRA1, MFGE8, POSTN and FN1. Connectivity-based signature reversal (LINCS L1000) was uninformative, returning cytotoxic and anti-proliferative perturbagens rather than anti-fibrotic candidates, an expected limitation for signatures dominated by structural matrix genes. We therefore nominate candidates mechanistically: the approved anti-fibrotics **nintedanib** (PDGFR/FGFR/VEGFR inhibitor) and **pirfenidone** (TGF-β1 modulator) act on the pathways upstream of the replicated program and are amenable to local intra-ligamentous delivery. We position these against existing prior art, distinguishing degenerative LF hypertrophy from post-surgical epidural fibrosis (for which both drugs have precedent), and release the analysis openly. This is a discovery-stage, in-silico hypothesis intended to motivate wet-lab evaluation; it is not preclinical validation.

1. Introduction

Lumbar spinal stenosis, a leading cause of disability and spine surgery in older adults, is driven in part by hypertrophy of the ligamentum flavum (LF): a chronic, degenerative thickening of the ligament that narrows the spinal canal. LF hypertrophy is fundamentally a fibrotic process — loss of elastic fibres, accumulation of type I/III collagen, fibroblast-to-myofibroblast conversion, and TGF-β1/Smad-driven matrix deposition. Despite a well-mapped mechanism, there is no approved drug that slows or reverses LF hypertrophy; management remains surgical decompression once stenosis is established.

Two obstacles have limited computational drug-repurposing approaches here. First, human single-cell data for LF is scarce. Second, disease-versus-control transcriptional contrasts derived from one small cohort frequently fail to reproduce in another. We address both by (i) building the disease signature as a within-tissue cell-state contrast rather than a case-versus-control contrast, and (ii) requiring the signature to reproduce in an independent cohort on a different platform before it is used. We then identify repurposing candidates by mechanism rather than by connectivity mapping, which we show is ill-suited to this signature, and we assess the resulting hypothesis against the existing patent and literature landscape.

2. Methods

2.1 Data

Single-cell discovery cohort: GSE294458 (human LF, 10x Genomics; one hypertrophic and one non-hypertrophic donor). Independent single-cell replication cohort: GSE267819 (human LF, 10x Genomics; three hypertrophic donors; Ham et al.). Independent bulk replication cohort: GSE113212 (human LF, Agilent SurePrint G3 microarray; four elderly/hypertrophic versus four young/non-hypertrophic samples). All are public; no unpublished or author-provided data were used.

2.2 Single-cell processing and signature definition

Standard quality control was applied, including an explicit hemoglobin-fraction filter to remove erythrocyte contamination, and hemoglobin and mitochondrial genes were excluded from feature selection. Cells were clustered and annotated to compartments (fibroblast, macrophage, endothelial, and others) by canonical markers, with an erythrocyte label to exclude residual red-blood-cell clusters. Within the fibroblast compartment, cells were sub-clustered and scored for a myofibroblast/activation program; the disease signature was defined as the differential expression between the highest-scoring (activated) and lowest-scoring (resting) fibroblast states, pooling both donors so the contrast reflects fibroblast state rather than the single-donor difference. The ranked signature was scrubbed of ribosomal, translation-factor, mitochondrial, hemoglobin and immunoglobulin genes, which are technical noise. Because only the activated (UP) pole reproduced (see Results), the signature was locked as UP-anchored.

2.3 Cross-platform replication

Replication was assessed in two independent cohorts. In the single-cell cohort (GSE267819), the same within-fibroblast activated-versus-resting contrast was computed, and the discovery signature was tested for concordance (fraction of signature genes up in activated fibroblasts) and for rank-enrichment within that cohort's activated ranking. In the bulk cohort (GSE113212), per-gene differential expression (hypertrophic versus young) was computed by Welch t-test with log2 fold-change, and the signature was tested by the fraction of genes with positive bulk fold-change and by rank-enrichment against the full bulk ranking (one-sided Mann-Whitney).

2.4 Candidate identification and prior-art assessment

Connectivity-based reversal was attempted with LINCS L1000 (L1000CDS2), including an approved-drug filter via the Broad Drug Repurposing Hub. As this proved uninformative (Results 3.3), candidates were instead nominated by matching approved anti-fibrotic mechanisms to the pathways upstream of the replicated program. The resulting hypothesis was assessed against issued and pending patents (Google Patents, USPTO, WIPO) and the peer-reviewed literature.

3. Results

3.1 An activated-fibroblast program in hypertrophic LF

The within-fibroblast activated-versus-resting contrast yielded a coherent matrix-depositing program: fibrillar collagens (COL1A1, COL1A2, COL3A1, COL5A2), small leucine-rich proteoglycans (LUM, ASPN, OGN, DCN), the matrix-remodeling genes HTRA1 and MFGE8, and the fibroblast-activation marker S100A4/FSP1. This is the effector program of fibroblast-to-myofibroblast activation and is consistent with the established histopathology of LF hypertrophy.

3.2 The signature replicates across independent cohorts and platforms

The signature reproduced in both independent cohorts (Figure 1). In the second single-cell cohort (GSE267819; 13,567 fibroblasts across three donors), the activated-versus-resting contrast recovered the same program: all 22 tested signature genes were up in activated fibroblasts (median log2 fold-change +1.6), with rank-enrichment $p = 9.4\times10^{-16}$, and the cohort's own top activated genes were dominated by fibrillar collagens, HTRA1, ASPN, MFGE8, DKK3 and FN1. In the bulk microarray cohort (GSE113212), the program reproduced despite bulk-tissue dilution and an age-based contrast: 18 of 20 genes showed positive fold-change (rank-enrichment $p = 1.5\times10^{-7}$), with the canonical fibrosis core elevated and mostly significant, including FN1 (+3.1), POSTN (+2.5), HTRA1 (+1.76), LUM (+1.89) and every fibrillar collagen. Replication across two independent single-cell cohorts and one bulk cohort on a different platform indicates the signature reflects LF disease biology rather than a single-cohort artifact. The resting (DOWN) pole did not reproduce, as expected, because its inflammatory/secretory genes rise with age and inflammation; it was therefore dropped.

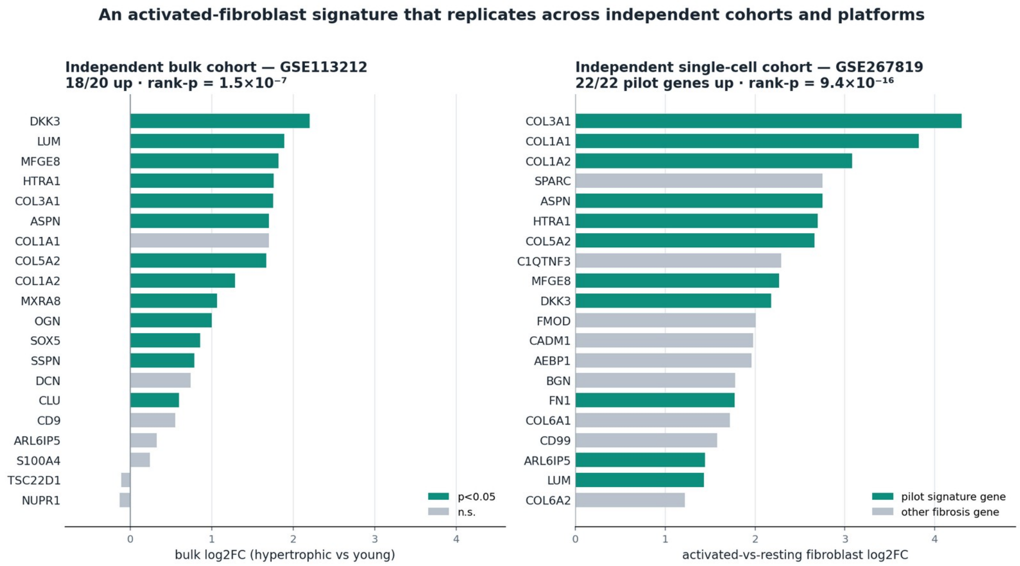

Figure 1. The activated-fibroblast signature replicates across independent cohorts. Left: pilot signature genes in independent bulk LF (GSE113212); bulk log2 fold-change, hypertrophic vs young; teal = $p < 0.05$; 18/20 up, rank-p = 1.5×10^{-7}. Right: top activated-fibroblast genes in an independent single-cell cohort (GSE267819); activated-vs-resting log2 fold-change; teal = pilot signature gene; all 22 pilot genes up, rank-p = 9.4×10^{-16}.

3.3 Connectivity mapping was uninformative

Signature-reversal querying of LINCS L1000 returned perturbagens dominated by cytotoxic and anti-proliferative agents (tubulin, HDAC, topoisomerase, Chk1, and PI3K/mTOR inhibitors) and chemical-probe tool compounds; after filtering to approved, non-oncology drugs, no credible anti-fibrotic candidate remained. This is expected: a signature dominated by structural matrix genes is nominally “reversed” by anything that halts secretory or proliferating cells, and the gentle, pleiotropic anti-fibrotics of therapeutic interest produce weak L1000 signatures that connectivity ranking disfavors. Connectivity mapping is therefore the wrong instrument for this problem, which motivates mechanism-based nomination.

3.4 Mechanism-matched candidates

Two approved anti-fibrotics map directly onto the replicated program. **Nintedanib** (approved for idiopathic pulmonary fibrosis, systemic-sclerosis–associated ILD, and progressive fibrosing ILD) inhibits

PDGFR, FGFR and VEGFR — receptors driving fibroblast activation, and, via VEGFR, the angiogenic coupling implicated in LF hypertrophy — all upstream of the collagen/matrix effector program observed here. **Pirfenidone** (approved for IPF) suppresses TGF-β1 and collagen synthesis, directly upstream of the replicated collagen program. Both are amenable to local intra-ligamentous or peri-ligamentous depot delivery, consistent with the localization principle for tissue-restricted anti-fibrotic action. We nominate nintedanib as the lead on mechanistic breadth, with pirfenidone as an in-class companion.

3.5 Prior-art positioning

No published study has tested nintedanib or pirfenidone in LF or LF hypertrophy, and no patent claims either drug for any spinal indication. The nearest art is *post-surgical epidural fibrosis*, where both drugs have precedent (a locally-delivered pirfenidone spinal hydrogel; rat laminectomy studies of both drugs). Epidural fibrosis is a distinct pathology — acute, iatrogenic scar in the epidural space — whereas LF hypertrophy is chronic degenerative thickening of the ligament body; the delivery target (intra-ligamentous) and disease context also differ. A pending application (US 2024/0026359, University of Pittsburgh) claims the LF-hypertrophy indication with local delivery but only for a microRNA (miR-29a) agent, not small molecules. The mechanism these drugs share with LF fibrosis is what makes them plausible, and also what limits the novelty of the concept.

4. Discussion

Two results are worth emphasizing. First, a disease-relevant fibroblast-activation signature can be recovered from a single, minimal single-cell LF dataset and shown to reproduce across platforms, provided it is built as a within-tissue cell-state contrast and validated by rank-based replication rather than trusted from one cohort. Second, this replicated signature points, by mechanism, to two approved, locally-deliverable anti-fibrotics as repurposing hypotheses for a disease with no approved pharmacotherapy.

The limitations are substantial and stated plainly. The single-cell evidence rests on one cohort of one donor per group; the replication cohort is bulk tissue with an age-based rather than pathology-graded contrast; and the entire analysis is computational, with illustrative rather than measured pharmacology. Connectivity mapping failed to nominate candidates, so the drug hypotheses are mechanistic inferences, not data-driven hits. Nintedanib and pirfenidone are the archetypal anti-fibrotics acting on the very pathway that defines the disease, so applying them to another fibrotic tissue is a rational but modest inductive step; the shared mechanism that makes them plausible also makes the concept unsurprising.

We release this openly rather than seeking patent protection. The therapeutic concept is mechanistically obvious and adjacent prior art exists in epidural fibrosis; a defensible claim would require specific formulation and efficacy data that a computational program cannot generate. Open publication establishes the finding, invites the wet-lab groups who can test it, and serves the translational goal better than a narrow filing. We hope the replicated signature and the candidate rationale are useful to laboratories positioned to evaluate local anti-fibrotic therapy for ligamentum flavum hypertrophy.

Data and code availability

Analyses use public data (GSE294458, GSE267819, GSE113212). Processing notebooks (single-cell signature, both replication analyses, and reversal analyses) are available in the companion repository, github.com/glenritschel/lf-hypertrophy-repurposing, deposited alongside this preprint.

Competing interests

None declared. No patent has been filed on this work; it is released openly.

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